IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

TRAFFIC THEORY AND QUEUING SYSTEMS

Monday, 29 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Mandic

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Special information for students

1. Erlang Loss formula recursive evaluation:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

2. Engset Loss formula recursive evaluation (for a fixed M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M-N+1)\alpha e_{N-1}}{N + (M-N+1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Kleinrock's conservation law:

$$\sum_{k=1}^{m} \rho_k E[W_k] = \left(\frac{\rho}{1-\rho}\right) E[R]$$

4. Expected residual time

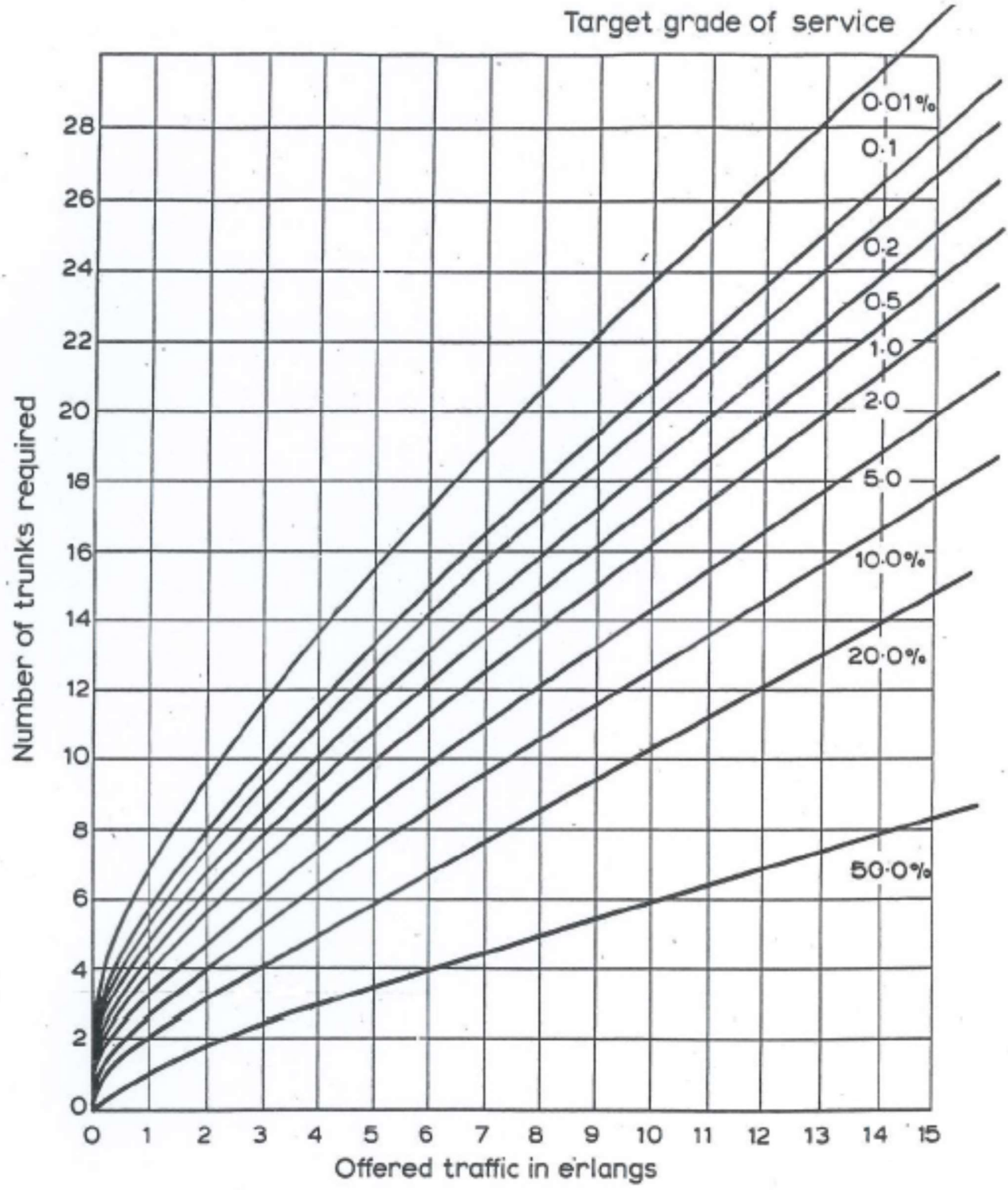
$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m} \lambda_k E[S_k^2]$$

5. Mean and variance of the Kosten overflow model (infinite overflow link $N = \infty$).

$$m = E[N_t] = \rho E_M(\rho)$$

$$v = Var(N_t) = m\left(1 - m + \frac{\rho}{M+m+1-\rho}\right)$$

6. Traffic capacity on basis of Erlang B formula (next page).



Traffic capacity on basis of Erlang B. formula.

The Questions

1. Birth/death process.

a) Consider the unbuffered system shown in **Fig. 1.1.** where:

- the number of sources, S, is many orders of magnitude larger than the number, N, of channels in the system.

- the arrival demand stream is a Poisson process with parameter $\boldsymbol{\lambda_S}$.

- the channel holding times are independent and exponential random variable with mean holding time $1/\mu$.

- the system enables full availability.

i) Sketch the state transition diagram of the Markov chain representing the number of busy channels on the link at time t. [2]

ii) Derive the transition rate parameters of the system described in a-i). Clearly explain your derivations. [2]

iii) Derive the steady state distribution for the system described in a-i). Clearly explain your derivations. [3]

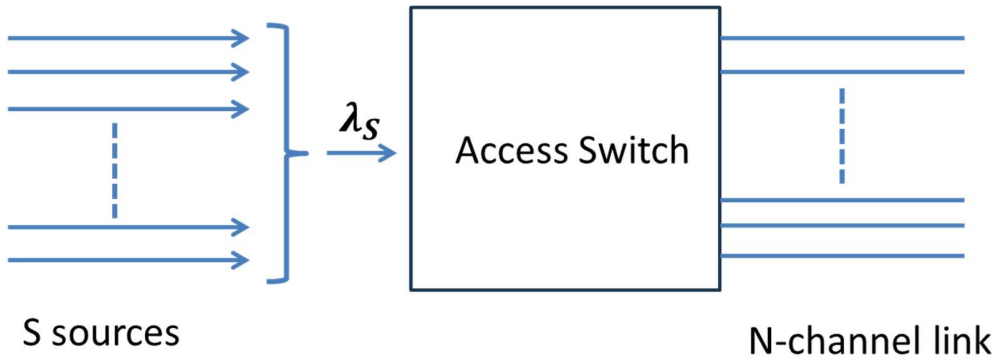


Figure 1.1.

b) Consider the unbuffered system shown in **Fig. 1.2.** where:

- the number, M, of sources which are Poisson streams with parameter λ acting independently, are of the same order of magnitude than the number of channels N.

- the channel holding times are independent and exponential random variable with mean holding time 1/μ.

- the system enables full availability.

Hint: that for this system if j channels are busy, then also j sources are busy.

i) Sketch the state transition diagram of the Markov chain representing the number of busy channels on the link at time t. [2]

ii) Derive the transition rate parameters of the system described in b-i). Clearly explain your derivations. [2]

iii) Derive the steady state distribution for the system described in b-i). Clearly explain your derivations. Assume $N \geq M$. [3]

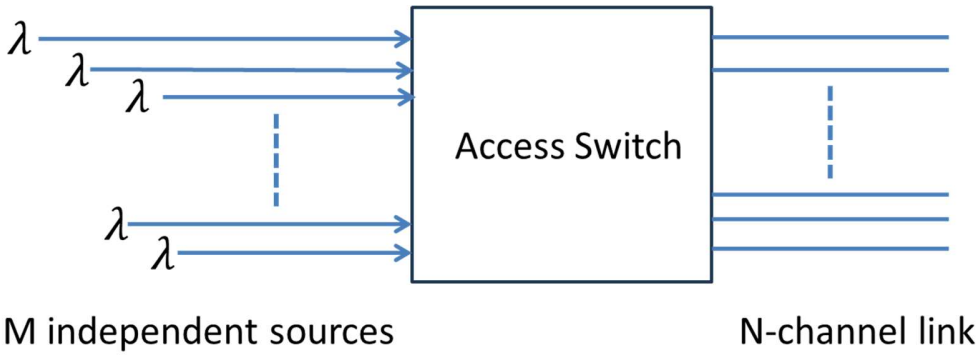


Figure 1.2.

c)

i) In the context of a Mean Value Analysis framework, briefly explain the relevance of tracking the residual time on a $M/G/1$ system. [2]

ii) Explain what are the advantages of considering an equivalent capacity function when multiplexing broadband traffic sources. [2]

iii) For a Poisson stream, derive the probability that there will be an arrival in the next interval of length Δt. [2]

2. Buffered systems and Markov modulated models.

a) For the system shown in **Fig. 2.1**

- the arrival demand stream is a Poisson process with parameter λ.

- the channel holding times are independent and exponential random variable with mean holding time $1/\mu$.
- the system has K servers available, and a buffer with unlimited capacity.

Assuming that the system enables full availability access, and the queueing discipline is first in first out ($FIFO$):

i) Sketch the Markov chain model of this system. Clearly identify all transition parameters. [3]

ii) Derive the probability that the system is not being utilised. [5]

iii) Derive the queue length distribution seen by demands arriving that find all K servers busy. [3]

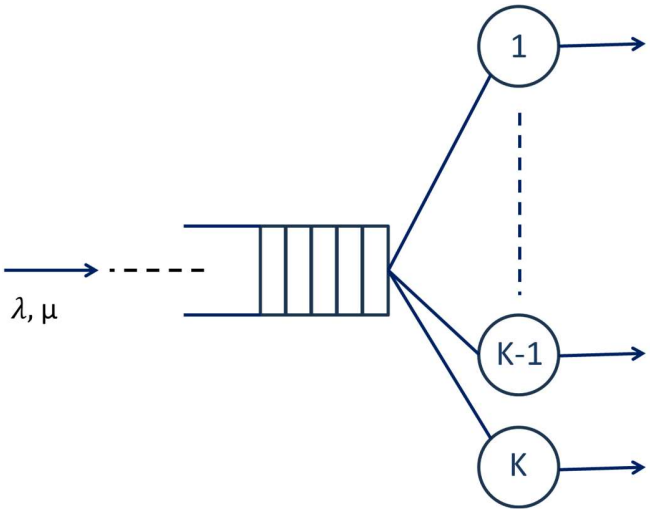


Figure 2.1

b) For the Markov modulated model shown in **Fig. 2.2**, it is known that:

- when the system is in state 3 it offers a demand stream that is Poisson distributed with parameter $\lambda_3 = 5.0$ demands/s.

- when the system is in state 1, the system is not offering any traffic, i.e. $\lambda_1 = 0.0$ demands/s.

i) Assume that it is possible to modulate a Poisson stream when in state 2 with parameter λ_2.

If the model is required to produce a mean rate of 3.2 demands/s in the long run, derive the value of λ_2 that will satisfy such design constraint. [4]

ii) Assume now that you can use an Interrupted Poisson process model (**Fig. 2.3**) with parameters $\alpha = 3$ and $\beta = 2$. Derive the parameter λ_1 of the modulated Poisson stream when the state is active, to produce a mean rate over the long run of 3.2 demands/s. [3]

iii) Discuss in your own words the advantages and disadvantages of the Markov Modulated Poisson process in b-i) versus the Interrupted Poisson process in b-ii). [2]

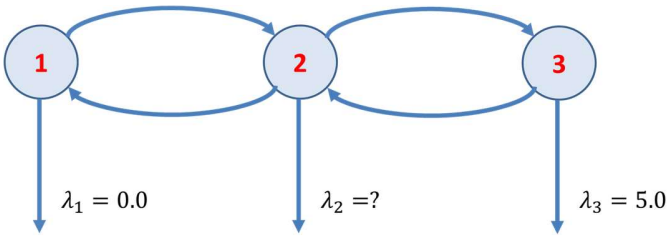


Figure 2.2.

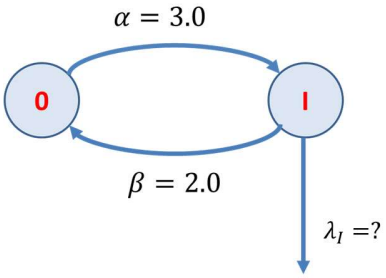


Figure 2.3.

3. Non-Markov queuing system.

a) Consider two Poisson arrival streams that are offered to a server (**Fig. 3.1.**):

- Poisson stream 1 has a mean arrival rate of $\lambda_1 = 225$ arrivals/s, and the packet size is $D_1 = 160$ bits.

- Poisson stream 2 has a mean arrival rate of $\lambda_2 = 75$ arrivals/s, and the packet size is $D_2 = 320$ bits.

i) Consider first, that the two Poisson arrival streams are statistically merged into one stream, and the aggregate stream offered to a server endowed with transmission rate of $C = 64$ Kbit/s. Assume also, a FIFO queue discipline.

Derive the mean arrival transit time, that is, the sum of the waiting time and the transmission time. [6]

ii) Consider next, a modified system that can implement ***non-pre-emptive priority*** to packets from stream 1:

- Derive the mean transit time for stream 1 packets. [3]

- Derive the mean transit time for stream 2 packets. [2]

- Derive the overall mean arrivals transit time. [3]

iii) Is this a work-conserving system?

Explicitly verify Kleinrock's conservation law. [6]



Figure 3.1.

4. Fluid flow model and Performability.

a) For an $N-$ source access buffer stochastic fluid model, shown in **Fig. 4.1.**, the probability that the buffer occupancy is less or equal to x with i sources active at time t is given by the expression: $F_i(t,x)$ for $0 \leq i \leq N$, where x is defined as the number of cells arriving during a talk spurt.

Hint: the following equivalents for this system are given:

One voice source generating V cells/s during a talk spurt of an average $1/\alpha$ s will increment x by V/α cells (unit of information) during a talk spurt.

Hence a system with capacity VC will have an equivalent capacity of αC.

Also, if there are i sources in talk spurt they all will be offering $i\alpha$ unit of information.

i) Derive the rate at which the buffer is filling/depleting. [2]

ii) Give the conditions for the buffer to be net depleting. [2]

iii) Derive: $\frac{\partial F_i(x,t)}{\partial t}$ for this system. [5]

iv) If in this system stationarity can hold, state the conditions for this to happen. [2]

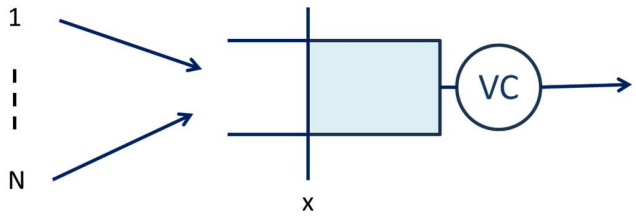


Figure 4.1.

b) The behaviour of a computer degradable system is shown in **Fig. 4.2.** where N is the number of operational computers.

The failure rate is $\lambda = 2.0$ failures/hour.

i) Describe in your own words what is the physical interpretation of the transition of the type $N(1-c)\lambda$. Assume $c = 0.999$. [2]

You are asked to analyse an $N(=2)$ redundant computer system.

ii) Add a transition to represent a full recovery of the system when it reaches the state $N = 0$. The transition rate is $\mu = 6.0$ repairs/hour. [2]

Now, you are required to augment the model with a reward structure: $R = [r_0, r_1, r_2]$. Answer the following questions:

iii) Derive the steady state reward delivered by the system constructed in b-ii). [3]

iv) Discuss a suitable reward structure to calculate the system availability. [2]

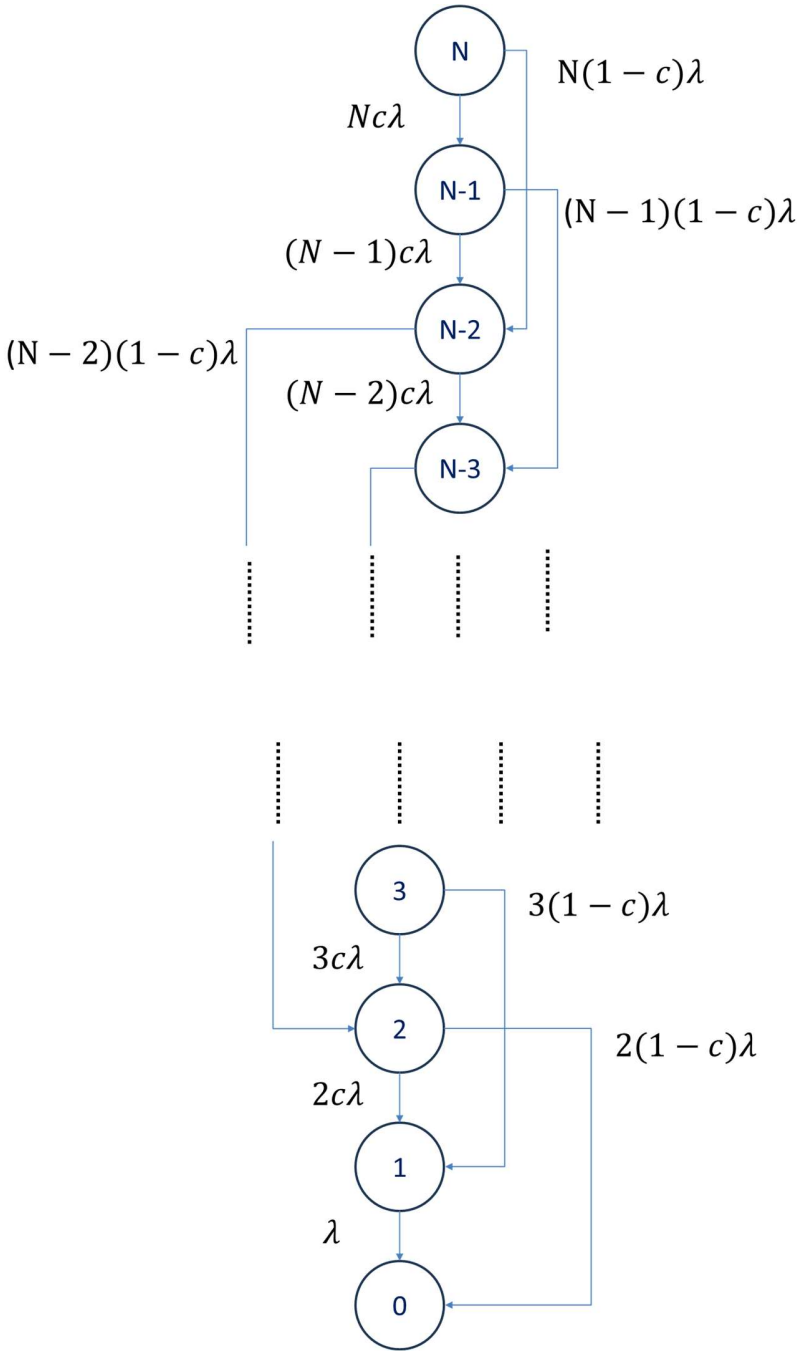

N
N(1 − c)λ
Ncλ
N-1
(N − 1)(1 − c)λ
(N − 1)cλ
N-2
(N − 2)(1 − c)λ
(N − 2)cλ
N-3
3
3(1 − c)λ
3cλ
2
2(1 − c)λ
2cλ
1
λ
0

Figure 4.2.